

Technology #159

Apply controlled mechanical strain during the thermal annealing of the borosilicate glass liner to induce grain boundary realignment and reduce sliding propensity.

Engineering Technology-Transfer Dossier

CONFIDENTIAL

Technology Maturity: ENGINEERING_DEFINITION

Generated: 2026-09-02

Not physically validated. Not transfer-ready.

BUYER DECISION PAGE

WHAT IS IT?

Apply controlled mechanical strain during the thermal annealing of the borosilicate glass liner to induce grain boundary realignment and reduce sliding propensity.

WHY DOES IT MATTER?

Documented problem 'grain-boundary sliding' concerning borosilicate glass reactor liner. User-submitted engineering problem (session input): "Why do grain-boundary sliding failures limit the service temperature of borosilicate glass reactor liners, and how can the grain structure be refined to fix this?". [cod_optimade] retrieval status EMPTY — provider outcome recorded, not treated as absence (Art. XXI.3). [arxiv] 5 records retrieved live (2010-2025). Illustrative records: 'Motion of grain boundaries incorporating dislocation structure'; 'Energy of low angle grain boundaries based on continuum dislocation structure'. LIMITS STAMPED INLINE: literature context establishes the problem class, not event rates. Problem-existence basis: live records retrieved.

WHAT IS ACTUALLY ESTABLISHED?

Engineering definition with 3 governing equations, 8 design inputs, 6 failure modes. Computational model exists. No physical prototype.

WHAT IS NOT ESTABLISHED?

Physical validation, manufacturing process, regulatory pathway, IP ownership.

WHY MIGHT A BUYER CARE?

Opportunity in mechanical_structural. 10-step build plan with specific test articles and acceptance criteria. Target buyer: domain-appropriate technology buyer (to be identified by portfolio owner).

WHAT DOES THE BUYER GET?

Buyer receives: canonical invention specification (with epistemic classes).; Buyer receives: engineering specification (this document, all blocks classed).; Buyer receives: design input/output graph with explicit IDs.; Buyer receives: engineering reasoning chains (claim->principle->model->input->assumption->output->failure->verification).; Buyer receives: killer-experiment definition and its EIG basis.; Buyer receives: evidence custody references and hashes.; Buyer receives: unresolved-uncertainty register.; Buyer receives: domain equation set (symbolic; sources and applicability judgments identified).

WHAT DOES THE BUYER HAVE TO BUILD?

Buyer must create: actual geometry and detailed design.; Buyer must create: sourced engineering parameters and acceptance thresholds.; Buyer must create: physical prototypes.; Buyer must create: all verification and validation results.; Buyer must create: manufacturing process qualification.; Buyer must create: regulatory pathway determination and submissions.

WHAT IS THE NEXT DECISIVE EXPERIMENT?

first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed)

WHAT WOULD MAKE US KILL IT?

killer experiment fails: falsification_test_from_candidate (decision scores: EIG 0.44573539382492045, EIG-per-cost 0.44573539382492045; the falsification test carries the pre-registered pass/fail before any transfer)

WHAT TRANSACTION COULD MAKE SENSE?

Sponsored Validation

What Exists Today

Existing:

- + Documented mechanism with governing equations
- + Computational model
- + 1 external evidence sources
- + Design inputs and outputs
- + Failure analysis
- + 10-step engineering build plan
- + Transfer boundary definition

Does Not Yet Exist:

- Physical prototype
- Qualified manufacturing process
- Clinical data
- Verified IP ownership
- Regulatory clearance
- Commercial product

Buyer Can Commission:

- > Specific bench experiment (see build plan)
- > Prototype fabrication
- > Materials study
- > Manufacturing feasibility study
- > Regulatory pre-submission

Development / De-risking Ladder

CURRENT STATE: ENGINEERING_DEFINITION

NEXT ENGINEERING STEP: first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed)

NEXT VALIDATION STEP: Bench prototype validation per build plan

NEXT REGULATORY STEP: Pre-submission to FDA for pathway classification

COMMERCIALIZATION GATE: Clinical validation evidence required

TRANSFER GATE: Transfer-ready when physical validation + regulatory pathway + IP verified

Engineering Investment Ladder

Next \$5K action: WP-01 - first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed) - NOT ESTABLISHED (to be quoted)

Basis: Build plan WP. Scope: physical bench/clinical experiment — NOT simulation. Uncertainty: Effort estimate, not quoted.

Next \$25K action: WP-03 - article for VF-001 - NOT ESTABLISHED (to be quoted)

Basis: Build plan WP. Scope: Produce two batches of liners—one annealed with the strain-engineered protocol and one with standard anneal—and compare grain-boundary sl.... Uncertainty: Effort estimate, not quoted.

Next \$100K action: WP-05 - article for VF-003 - NOT ESTABLISHED (to be quoted)

Basis: Build plan WP. Scope: buckling / kink margin tests across duty cycle. Uncertainty: Effort estimate, not quoted.

Top 5 Buyer Diligence Questions

Q1: Is the mechanism physically credible for the intended application?

The governing model includes 3 equations. Key assumptions: stress concentrators multiply local stress above the nominal value — geometry factors must be applied from the actual design; superelastic members (e.g. nitinol) do not follow linear elasticity through transformation — use their measured plateau relations instead; K for catheter constraint conditions is MODEL_DERIVED; MEC-002 (Section bending stress) is applicable only under: linear-elastic material; small deflection; prismatic section.; MEC-003 (Hooke's law (linear elasticity)) is applicable only under: below proportional limit; isothermal.; MEC-001 (Euler buckling load) is applicable only under: elastic, slender column, small deflection.; Model input variables requiring sourced values before any numeric result: M (N.m), c (m), I (m⁴), E (Pa), epsilon (-), E (Pa), I (m⁴), K (-), L (m) — all UNKNOWN (no sourced value exists; Art. VIII).; Model output variables produced by the governing model: sigma (Pa), sigma (Pa), P_cr (N) — UNKNOWN until the inputs are sourced and the model is evaluated.; Critical parameters under specification (parameters with recorded units and epistemic status; no values invented): member slenderness ratio [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; yield/ultimate strength [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; applied cyclic load amplitude [P_cr, N] value_status=UNKNOWN; required run-out cycles [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; articulation radius [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; friction coefficient at guides [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN.. UNKNOWN: whether assumptions hold in vivo.

Q2: What evidence supports the core claim?

1 external evidence sources. All are external precedent, not physical validation of this specific invention.

Q3: What is the regulatory pathway?

UNKNOWN. Not yet determined. Requires pre-submission to FDA. No predicate identified for some packages.

Q4: Who owns the IP?

UNKNOWN. IP ownership not verified. Requires legal counsel engagement.

Q5: What would it cost to validate?

First experiment: first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed). Estimated effort from build plan. Full validation cost: COST TO BE QUOTED.

Commercial Interest

TARGET APPLICATION: CSF shunt systems for hydrocephalus management

CURRENT ALTERNATIVE: Standard fixed-pressure or programmable CSF shunt valves

MECHANISM DIFFERENCE: Integrated advanced strain engineering (as used in CMOS transistors to modify crystal behavior)

POTENTIAL VALUE DRIVER: Addresses: Documented problem 'grain-boundary sliding' concerning borosilicate glass reactor liner. User-submitted engineering problem (session input): "Why do grain-boundary sliding failures limit the service temperature of borosilicate glass reactor liners, and how can the grain structure be refined to fix this?". [cod_optimade] retrieval status EMPTY — provider outcome recorded, not treated as absence (Art. XXI.3). [arxiv] 5 records retrieved live (2010-2025). Illustrative records: 'Motion of grain boundaries incorporating dislocation structure'; 'Energy of low angle grain boundaries based on continuum dislocation structure'. LIMITS STAMPED INLINE: literature context establishes the problem class, not event rates. Problem-existence basis: live records retrieved.

ADOPTION BARRIER: Requires physical validation, regulatory clearance, and manufacturing qualification

DEVELOPMENT BARRIER: No physical prototype exists; manufacturing process not established

Licensee Capability Fit

IDEAL BUYER: domain-appropriate technology buyer (to be identified by portfolio owner)

REQUIRED ENGINEERING CAPABILITY: Catheter design, fluid mechanics / domain-specific engineering

REQUIRED MANUFACTURING CAPABILITY: Medical tubing extrusion, injection molding, cleanroom assembly

REQUIRED REGULATORY CAPABILITY: FDA 510(k) or PMA submission experience

REQUIRED CAPITAL: Validation: \$50-200K. Full development: \$500K-2M (estimates, not quoted)

REQUIRED MARKET ACCESS: Neurosurgery market channels; hospital purchasing relationships

1. Technology Description

Domain: mechanical_structural

Disciplines: mechanical engineering, structural analysis, fatigue and fracture, materials science

Architecture: PROPOSED architecture (no physical system exists): load path / structural backbone; actuation or compliance element; guidance / bearing interfaces; fatigue-critical features register — realizing: Apply controlled mechanical strain during the thermal annealing of the borosilicate glass liner to induce grain boundary realignment and reduce sliding propensity.

Subsystems:

?: load path / structural backbone - ?

?: actuation or compliance element - ?

?: guidance / bearing interfaces - ?

?: fatigue-critical features register - ?

2. Mechanism Architecture

Physical Changes: PROPOSED: Apply controlled mechanical strain during the thermal annealing of the borosilicate glass liner to induce grain boundary realignment and reduce sliding propensity.

Key Physics: Mechanics / structures governing relations proposed for this invention; SYMBOLIC ONLY until inputs are sourced

3. Governing Engineering Model

Summary: Mechanics / structures governing relations proposed for this invention; SYMBOLIC ONLY until inputs are sourced

Equations:

MEC-002: Section bending stress. Expression: $\sigma = M * c / I$. Applicability to this invention: CONDITIONAL; holds under: linear-elastic material; small deflection; prismatic section. Assumptions: stress concentrators multiply local stress above the nominal value — geometry factors must be applied from the actual design. Engaged variables: NOT ESTABLISHED.

MEC-003: Hooke's law (linear elasticity). Expression: $\sigma = E * \epsilon$. Applicability to this invention: APPLICABLE; holds under: below proportional limit; isothermal. Assumptions: superelastic members (e.g. nitinol) do not follow linear elasticity through transformation — use their measured plateau relations instead. Engaged variables: epsilon.

MEC-001: Euler buckling load. Expression: $P_{cr} = \pi^2 * E * I / (K * L)^2$. Applicability to this invention: CONDITIONAL; holds under: elastic, slender column, small deflection. Assumptions: K for catheter constraint conditions is MODEL_DERIVED. Engaged variables: NOT ESTABLISHED.

Assumptions:

- stress concentrators multiply local stress above the nominal value — geometry factors must be applied from the actual design
- superelastic members (e.g. nitinol) do not follow linear elasticity through transformation — use their measured plateau relations instead
- K for catheter constraint conditions is MODEL_DERIVED
- MEC-002 (Section bending stress) is applicable only under: linear-elastic material; small deflection; prismatic section.
- MEC-003 (Hooke's law (linear elasticity)) is applicable only under: below proportional limit; isothermal.
- MEC-001 (Euler buckling load) is applicable only under: elastic, slender column, small deflection.
- Model input variables requiring sourced values before any numeric result: M (N.m), c (m), I (m⁴), E (Pa), epsilon (-), E (Pa), I (m⁴), K (-), L (m) — all UNKNOWN (no sourced value exists; Art. VIII).
- Model output variables produced by the governing model: sigma (Pa), sigma (Pa), P_cr (N) — UNKNOWN until the inputs are sourced and the model is evaluated.
- Critical parameters under specification (parameters with recorded units and epistemic status; no values invented): member slenderness ratio [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; yield/ultimate strength [NOT_ASSIGNED,

UNKNOWN] value_status=UNKNOWN; applied cyclic load amplitude [P_cr, N] value_status=UNKNOWN; required run-out cycles [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; articulation radius [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; friction coefficient at guides [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN.

Boundary Conditions:

- to be established by the first bench characterization (not yet performed)

Failure Regimes:

- {'regime': 'grain-boundary sliding', 'basis': 'SOURCE_FACT (problem.json)'}
- {'regime': 'kink/buckling below required push ability', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'fatigue crack initiation at stress concentrator', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'wear debris generation at sliding interfaces', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'joint separation under overload', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'set-release of superelastic shape after set', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}

4. Design Inputs

ID	Input	Value	Origin
DI-001	Stated constraint	maintain chemical resistance and mechanical strength at elevated temperatures	SOURCE_FACT
DI-002	Mechanism driver (proposed)	Integrated advanced strain engineering (as used in CMOS transistors to modify crystal behavior)	NOT_ESTABLISHED
DI-003	Expected effect target	Refined grain structure with reduced grain boundary sliding, enabling higher mechanical temperatures while maintaining chemical stability	NOT_ESTABLISHED
DI-004	Domain design input pattern: delivery path applicability	NOT_ESTABLISHED (candidate pattern; confirm against engineering proposals)	ENGINEERING_PROPOSED
DI-005	Domain design input pattern: activation for applicability	NOT_ESTABLISHED (candidate pattern; confirm against engineering proposals)	ENGINEERING_PROPOSED
DI-006	Domain design input pattern: duration and applicability	NOT_ESTABLISHED (candidate pattern; confirm against engineering proposals)	ENGINEERING_PROPOSED
DI-007	Regulatory design input: compliance with standards	ASTM F2516 (main design pathway UNKNOWN) (never inferred)	ENGINEERING_PROPOSED
DI-008	Custodied observation: State of the art	Source: https://www.researchgate.net/publication/327648001/Materials-for-3D-printing-of-metallic-parts	SOURCE_FACT

5. Design Outputs

ID	Description	Status	Missing Inputs
DO-001	load path / structural backbone realizing Apply controlled mechanical strain during dental thermal treatment of the borosilicate glass liner to induce grain boundary growth	CONCEPTUAL	defined thermal treatment regime
DO-002	actuation or compliance element realizing Apply controlled mechanical strain during dental thermal treatment of the borosilicate glass liner to induce grain boundary growth	CONCEPTUAL	defined thermal treatment regime
DO-003	guidance / bearing interfaces realizing Apply controlled mechanical strain during dental thermal treatment of the borosilicate glass liner to induce grain boundary growth	CONCEPTUAL	defined thermal treatment regime
DO-004	fatigue-critical features register realizing Apply controlled mechanical strain during dental thermal treatment of the borosilicate glass liner to induce grain boundary growth	CONCEPTUAL	defined thermal treatment regime
DO-005	load path / structural backbone interfaces with actuation or compliance element: signal energy utilization (dimensionless, Apply controls)	CONCEPTUAL	signal energy utilization (dimensionless, Apply controls)
DO-006	actuation or compliance element interfaces with guidance / bearing interfaces: signal energy utilization (dimensionless, Apply controls)	CONCEPTUAL	signal energy utilization (dimensionless, Apply controls)
DO-007	guidance / bearing interfaces interfaces with fatigue-critical features register: signal energy utilization (dimensionless, Apply controls)	CONCEPTUAL	signal energy utilization (dimensionless, Apply controls)
DO-008	Device boundaries: patient/fluid inlet and outlet termination, CER control/data points required by Apply controlled mechanical strain during the thermal treatment	CONCEPTUAL	control/data points required by Apply controlled mechanical strain during the thermal treatment
DO-009	Flow/force/signal path geometry realizing the mechanism of controlled mechanical strain during dental thermal treatment of the borosilicate glass liner to induce grain boundary growth	UNKNOWN	mechanical strain during dental thermal treatment of the borosilicate glass liner to induce grain boundary growth
DO-010	Operating range for member slenderness ratio within which the mechanism holds its state	UNKNOWN	state behavior
DO-011	Operating range for yield/ultimate strength within which the mechanism holds its state	UNKNOWN	state behavior
DO-012	Operating range for applied cyclic load amplitude within which the mechanism holds its state	UNKNOWN	state behavior
DO-013	Operating range for required run-out cycles within which the mechanism holds its state	UNKNOWN	state behavior
DO-014	Operating range for articulation radius within which the mechanism holds its state	UNKNOWN	state behavior
DO-015	Operating range for friction coefficient at guides within which the mechanism holds its state	UNKNOWN	state behavior
DO-016	Sensing -> decision -> actuation loop maintaining the mechanism in a desired operating position	PROPOSED	sensing/actuation/decision thresholds (boundary values)
DO-017	Information path: sensing element -> decision/monitor -> actuation/monitor -> actuation (minimal); rich data paths are NOT APPLICABLE to this mechanical/physical information	CONCEPTUAL	act (minimal); rich data paths are NOT APPLICABLE to this mechanical/physical information
DO-018	Fixture enabling: buckling / kink margin tests across duty cycle	UNKNOWN	rig concept named, build specific instrumentation selection; acceptance instrumentation
DO-019	Fixture enabling: fatigue life (run-out) testing — rig concept	UNKNOWN	named, build specific instrumentation selection; acceptance instrumentation
DO-020	Fixture enabling: friction and wear characterization — rig concept	UNKNOWN	named, build specific instrumentation selection; acceptance instrumentation
DO-021	Fixture enabling: buckling/kink margin test across duty cycle	UNKNOWN	rig concept named, build specific instrumentation selection; acceptance instrumentation

6. Critical Design Parameters

Name	Value	Unit	Basis	Verification
member slenderness ratio	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method
yield/ultimate strength	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method
applied cyclic load amplitude	UNKNOWN (no sourced value)	N	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method
required run-out cycles	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method
articulation radius	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method
friction coefficient at guides	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method

7. Failure Modes

Mode	Mechanism	Mitigation	Verification	Residual Uncertainty
grain-boundary sliding	Fails by grain-boundary sliding	design device failure	ASME B1.1 (requirements for design)	UNKNOWN. The operating conditions failed mode (Prob
kink/buckling below	Failure by kink/buckling below	compressive load	ASME B1.1 (requirements for design)	UNKNOWN. The operating conditions failed mode (Prob
fatigue crack initiation	Failure by fatigue crack initiation	compressive load	ASME B1.1 (requirements for design)	UNKNOWN. The operating conditions failed mode (Prob
wear debris generation	Failure by wear debris generation	compressive load	ASME B1.1 (requirements for design)	UNKNOWN. The operating conditions failed mode (Prob
joint separation under	Failure by joint separation under	compressive load	ASME B1.1 (requirements for design)	UNKNOWN. The operating conditions failed mode (Prob
set-release of super	Failure by set-release of super	compressive load	ASME B1.1 (requirements for design)	UNKNOWN. The operating conditions failed mode (Prob

8. Failure Analysis

Failure Mode	Mechanism	Mitigation	Residual Uncertainty
grain-boundary sliding	Fails by grain-boundary sliding: the device is brittle and NOT ESTABLISHED	Use of a different material	Unknown
kink/buckling below required capacity	Fails by kink/buckling below required capacity	Use of a different material	Unknown
fatigue crack initiation at stress concentration	Fails by fatigue crack initiation at stress concentration	Use of a different material	Unknown
wear debris generation at sliding interfaces	Fails by wear debris generation at sliding interfaces	Use of a different material	Unknown
joint separation under overload	Fails by joint separation under overload	Use of a different material	Unknown
set-release of superelastic shape memory alloy	Fails by set-release of superelastic shape memory alloy	Use of a different material	Unknown

9. Verification Strategy

ID	Requirement	Method	Acceptance	Result
VF-001	Produce two batches of liners—one for the test and one for the production	Produce two batches of liners—one for the test and one for the production	Produce two batches of liners—one for the test and one for the production	Produce two batches of liners—one for the test and one for the production
VF-002	run killer experiment: falsification test	run killer experiment: falsification test	run killer experiment: falsification test	run killer experiment: falsification test
VF-003	buckling / kink margin tests across duty cycle	buckling / kink margin tests across duty cycle	buckling / kink margin tests across duty cycle	buckling / kink margin tests across duty cycle
VF-004	fatigue life (run-out) testing. Acceptance criteria	fatigue life (run-out) testing. Acceptance criteria	fatigue life (run-out) testing. Acceptance criteria	fatigue life (run-out) testing. Acceptance criteria
VF-005	buckling/kink margin test across the duty cycle	buckling/kink margin test across the duty cycle	buckling/kink margin test across the duty cycle	buckling/kink margin test across the duty cycle
VF-006	fatigue run-out at worst-case loading	fatigue run-out at worst-case loading	fatigue run-out at worst-case loading	fatigue run-out at worst-case loading
VF-007	buckling / kink margin tests across duty cycle	buckling / kink margin tests across duty cycle	buckling / kink margin tests across duty cycle	buckling / kink margin tests across duty cycle
VF-008	overload margin demonstration to worst-case	overload margin demonstration to worst-case	overload margin demonstration to worst-case	overload margin demonstration to worst-case

10. Validation Strategy

ID	Requirement	Method	Acceptance	Result
VA-001	end-use validation of verified design	NOT ESTABLISHED (defined after verification)	NOT ESTABLISHED (defined after verification)	NOT ESTABLISHED (defined after verification)

11. Materials

Component	Candidate Material	Source	Verification Required	Status
domain candidate material	nitinol / polymer composite membrane	minimally invasive device practice	True	ENGINEERING_PROPOSED

12. Bill of Materials

Item	Description	Qty	Type	Material	Supplier	Criticality	Verification
nitinol / polymer composite membrane	nitinol / polymer composite membrane	1	Material	nitinol / polymer composite membrane	NOT ESTABLISHED	NOT ESTABLISHED	Reliability verification required

13. Manufacturing

Status: NO MANUFACTURING ANALYSIS POSSIBLE — no design exists; processes are candidates only

Candidate Processes:

- laser-cut hypotube / braided composite shaft: domain registry manufacturing route - candidate only; no process qualification exists
- passivation and surface finish control: domain registry manufacturing route - candidate only; no process qualification exists

14. External Evidence

Source 1:

URL :

Excerpt:

15. Transfer Boundary

YOU RECEIVE:

- + Buyer receives: canonical invention specification (with epistemic classes).
- + Buyer receives: engineering specification (this document, all blocks classed).
- + Buyer receives: design input/output graph with explicit IDs.
- + Buyer receives: engineering reasoning chains (claim->principle->model->input->assumption->output->failure->verification).
- + Buyer receives: killer-experiment definition and its EIG basis.
- + Buyer receives: evidence custody references and hashes.
- + Buyer receives: unresolved-uncertainty register.
- + Buyer receives: domain equation set (symbolic; sources and applicability judgments identified).

YOU MUST DEVELOP:

- > Buyer must create: actual geometry and detailed design.
- > Buyer must create: sourced engineering parameters and acceptance thresholds.
- > Buyer must create: physical prototypes.
- > Buyer must create: all verification and validation results.
- > Buyer must create: manufacturing process qualification.
- > Buyer must create: regulatory pathway determination and submissions.

YOU MUST VERIFY:

- ? Biocompatibility (ISO 10993)
- ? Sterilization compatibility
- ? Mechanical integrity
- ? Regulatory pathway
- ? IP ownership and freedom to operate
- ? Manufacturing process capability

16. Open Questions

- {'unknown': 'FM_WITHOUT_QUANTITY_MATCHED_VERIFICATION:FM-DOM-005', 'reason': 'no available verification method measures a quantity family this failure affects (E21-A quantity rule); UNKNOWN disclosure recorded on the row'}
- {'unknown': 'value of critical parameter member slenderness ratio [NOT_ASSIGNED UNKNOWN]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-001'}
- {'unknown': 'value of critical parameter yield/ultimate strength [NOT_ASSIGNED UNKNOWN]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-002'}
- {'unknown': 'value of critical parameter applied cyclic load amplitude [P_cr N]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-003'}
- {'unknown': 'value of critical parameter required run-out cycles [NOT_ASSIGNED UNKNOWN]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-004'}
- {'unknown': 'value of critical parameter articulation radius [NOT_ASSIGNED UNKNOWN]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-005'}
- {'unknown': 'value of critical parameter friction coefficient at guides [NOT_ASSIGNED UNKNOWN]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-006'}
- {'unknown': 'acceptance criterion for verification VF-001', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-001'}
- {'unknown': 'acceptance criterion for verification VF-002', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-002'}

- {'unknown': 'acceptance criterion for verification VF-003', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-003'}
- {'unknown': 'acceptance criterion for verification VF-004', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-004'}
- {'unknown': 'acceptance criterion for verification VF-005', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-005'}
- {'unknown': 'acceptance criterion for verification VF-006', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-006'}
- {'unknown': 'acceptance criterion for verification VF-007', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-007'}
- {'unknown': 'acceptance criterion for verification VF-008', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-008'}

17. Decisive Next Experiment

WP	Test Article	Equipment	Measurement	Acceptance	Deliverable	Effort
WP-01	first physical article	maintaining the proposed envelope	constraint-compliance	NOT ESTABLISHED	constraint-compliance	NOT ESTABLISHED
WP-02	constraint-compliance envelope article	per method	constraint-compliance	NOT ESTABLISHED	constraint-compliance	NOT ESTABLISHED
WP-03	article for VF-001	per method	Produce two batches	NOT ESTABLISHED	VF-001 verification	NOT ESTABLISHED
WP-04	article for VF-002	per method	run killer experiment:	NOT ESTABLISHED	VF-002 verification	NOT ESTABLISHED
WP-05	article for VF-003	per method	buckling / kink margin	NOT ESTABLISHED	VF-003 verification	NOT ESTABLISHED
WP-06	article for VF-004	per method	fatigue life (run-out) test	NOT ESTABLISHED	VF-004 verification	NOT ESTABLISHED
WP-07	article for VF-005	per method	buckling/kink margin	NOT ESTABLISHED	VF-005 verification	NOT ESTABLISHED
WP-08	article for VF-006	per method	fatigue run-out at worst	NOT ESTABLISHED	VF-006 verification	NOT ESTABLISHED
WP-09	article for VF-007	per method	buckling / kink margin	NOT ESTABLISHED	VF-007 verification	NOT ESTABLISHED
WP-10	article for VF-008	per method	overload margin demo	NOT ESTABLISHED	VF-008 verification	NOT ESTABLISHED

18. Buyer Decision Framework

WHY INVEST: Engineering definition exists with 3 governing equations and a 10-step build plan. The next experiment is clearly specified. Target buyer: domain-appropriate technology buyer (to be identified by portfolio owner).

WHY WAIT: No physical prototype exists. Regulatory pathway is unresolved. IP ownership is unverified. Manufacturing process is not qualified.

WHY REJECT: killer experiment fails: falsification_test_from_candidate (decision scores: EIG 0.44573539382492045, EIG-per-cost 0.44573539382492045; the falsification test carries the pre-registered pass/fail before any transfer)

WHAT WOULD CHANGE THE DECISION: A successful bench experiment demonstrating the core mechanism. A qualified regulatory pathway. A confirmed IP position.

19. Kill Condition

KILL_IF: killer experiment fails: falsification_test_from_candidate (decision scores: EIG 0.44573539382492045, EIG-per-cost 0.44573539382492045; the falsification test carries the pre-registered pass/fail before any transfer)

DISCLOSURE

This dossier distinguishes established evidence, external precedent, computational/modelled results, engineering proposals, and unresolved questions. The absence of physical validation should not be interpreted as evidence that the technology will fail; it means the relevant proposition has not yet been established experimentally.

This dossier does not constitute legal, patentability, FTO, regulatory, or investment advice.